

Detail Project Proposal for COMP7705 Project

Project Title

Detection and Localization of Spoofing and Jamming Signals: A Deep Learning-Driven Framework for Robust Signal Authentication

Mentor

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Abstract

The Global Positioning System (GPS) under the Global Navigation Satellite Systems (GNSS), plays an important role in the calculation on positioning, navigation, and timing (PNT) during the extensive application of Unmanned Aerial Vehicle (UAV) for computation on the altitude and flight path. When GPS receiver of the UAV fails to screen out spoofing and jamming signals, those misleading data interferes with the calculation on PNT and jeopardies system integrity. Therefore, it is crucial to detect and filter those signals from genuine signals to secure operation accuracy. Machine learning and Deep learning were introduced to mitigate such threats and this project aims to enhance existing strategies by training a simplified model for UAV platform by providing precise PNT for the real-world data, converting the signal into a spectral graph, while distinguishing malicious signals. Other data enhancement methods will also be explored and the effectiveness of the proposed adjustment would be measured through comparison in accuracy of current techniques.

Keywords: Machine Learning, Computer Vision, CNN, GNSS, GPS, Spoofing Detection, Jamming Detection, UAV, Navigation

1.Introduction

1.1 Background – Mechanism of Spoofing and Jamming

Global Navigation Satellite Systems (GNSS) are the dominant positioning systems in modern society, with major global service providers including Galileo, BeiDou, and GPS. The applications of GNSS are widespread, ranging from vehicle navigation to UAV positioning.

The working principle of GPS is based on satellite navigation technology, where satellites continuously transmit radio signals containing time and location information to the ground. A GPS receiver determines the user's position, velocity, and time by receiving signals from multiple satellites. The critical factor in GPS functionality is that if the receiver fails to

correctly receive GPS signals, the entire positioning system becomes ineffective. GPS signal interference is primarily categorized into two types: jamming and spoofing.

While inertial navigation systems (INS) [11] can mitigate this issue in civilian applications, in military scenarios—particularly missile guidance—there is an urgent need for a rapid and accurate method to distinguish legitimate GPS signals. This distinction is crucial for the precision and feasibility of missile guidance. Traditional signal processing algorithms can address this problem [10]; however, there is still room for optimization in terms of accuracy and processing speed.

With the rise of artificial intelligence and machine learning, the use of machine learning and deep learning has provided us with new ideas for distinguishing real GPS signals.

1.2 Research Objectives

- Machine learning and deep learning techniques and algorithm will be further studied to enhanced accuracy, detection ability for mitigating spoofing and jamming.
- Visualize the effectiveness in processing of data with an interface.

2.Literature Review

2.1 Mitigation for Spoofing

Existing studies related to detection on spoofing includes artificial neural networks (ANNs) classification with noise ratio [3], Linear Regression and Long Short-Term Memory (LSTM) [4], algorithms for Support Vector Machine (SVM) [5] with windows and time width, DeepSIM [6] through comparison with image processing between previous GPS image and receiving image, ten commonly used machine learning models were studied to derive two approaches for selection [7], 13 features were involved when tree based models were compared [8], combination of SVM and K-fold [9], that models with different techniques and algorithm were studied to accuracy, detection ability for mitigating spoofing.

2.2 Detecting Jamming

Czech et al. (2018) proposed an innovative deep learning framework [10] that uses CNN to extract local spatial features of spectrograms (such as pulse shapes), and then captures contextual dependencies of time series (such as periodicity of interference signals) through LSTM, ultimately achieving classification of interference types such as satellite signals and radar pulses. The cascaded architecture of CNN $\rightarrow$ LSTM has successfully achieved the extraction of spatiotemporal features, but there are still significant limitations in the feature interaction mechanism. The model only uses a simple feature transfer method, directly inputting the features extracted by CNN into LSTM, without exploring effective fusion of multi-scale features or introducing attention mechanism to achieve adaptive weighting of features, which to some extent limits the performance improvement space of the model.

3. Proposed Methodology

3.1 Data Source

Source of data is originated from open source data that there are three types of datasets for experimentation on detection, two for spoofing, one for jamming:

1. Spoofing. Data collected from legitimate GPS receiver on a vehicle which operated by simulating dynamic behavior during UAS operation. A total of 13 features for classification and identification of GPS Signal for training the model to distinguish between three types of GPS spoofing attacks.

2. Jamming. GNSS signal jamming spectrogram image dataset were processed by machine and deep learning upon being collected from Raw In-phase and Quadrature (IQ) data for testing GPS receivers on six classes of jamming attacks.

3.2 Tools

The implementation pipeline integrates the following tools and infrastructure:

Software Stack

Core Programming Languages: 1. Python: Primary language for machine/deep learning model development (using frameworks PyTorch/TensorFlow), data preprocessing (Pandas/NumPy), and visualization (Matplotlib/Seaborn). 2. MATLAB: Supplementary tool for signal processing-specific analyses (e.g., spectral filtering, wavelet transforms).

Web Interface: An interactive web dashboard will be built using HTML/CSS for frontend rendering, JavaScript (Plotly.js/D3.js) for dynamic visualization, and Flask as the backend framework to enable real-time data interpretation.

Hardware Infrastructure

Local GPU Acceleration: Primary training will utilize local NVIDIA GPUs (e.g., RTX 3090 with 24GB VRAM) via CUDA-accelerated libraries (cuDNN), prioritized for rapid prototyping.

Cloud-based Scaling: For large-scale experiments or distributed training, cloud GPU instances (AWS EC2 p3.8xlarge or Google Cloud A2 VMs) will be deployed, leveraging Kubernetes for workload orchestration.

3.3 Data Processing

3.3.1 Data Argumentation

Before training and testing the models with the public signal dataset, by conducting data strengthening or argumentation, image processing will be applied to convert signal data into spectrogram images to visualize the captured information of the amplitude and phase concerning the received signal over time. Short-Time Fourier Transform (STFT) would be one of the methods to achieve a time-frequency transformation for better illustration on how different types of jamming attack were identified and classified. The spectral features in the transformed image data facilitate classification of the jamming signals when input for machine and deep learning.

3.3.2 Data Enrichment and Balancing

Data enhancement will be performed to enlarge and increase sizes of existing datasets, apart from the 13 features, latitude, longitude and altitude will be added and beefed up in the existing 13 features of spoofing that other demographic and geographical information will be

explored for inclusion to increase predictability of the model. During initial classification, classes were observed to be imbalanced that oversampling and under sampling will be applied.

3.3.3 Filter and Noise Reduction

To further enhance data before inputting the data for training the models, filtering and noise reduction will be performed to increase reliability and performance.

3.3.4 Detection Model Setup

To compare and identify the optimal approach, multiple deep learning models will be employed for spectral data processing and image classification. Specifically, three advanced architectures will be implemented:

1. A 34-layer Residual Network (ResNet-34), leveraging skip connections to address gradient vanishing in deep networks;
2. A Convolutional Recurrent Neural Network (CRNN), combining convolutional layers for spatial feature extraction with recurrent layers (e.g., LSTM) for sequential pattern modeling;
3. A Vision Transformer (ViT), utilizing self-attention mechanisms to capture global dependencies in spectral images.

The model optimization will follow an iterative procedure:

1. Architecture refinement: Adjusting layer depth, kernel size, or attention heads (for ViT);
2. Hyperparameter tuning: Optimizing learning rate, weight decay, and dropout ratios;
3. Weight matrix fine-tuning: Transfer learning-based adaptation of pre-trained parameters.

The final model will be selected based on its performance on a held-out validation set, evaluated through metrics including accuracy, precision-recall balance, and computational efficiency.

4. Milestones

	Task	Estimated Completion Time	Estimated No. of Learning Hours
1	Collection of locations of airplane and cargo ships	2025-02-01	40
2	Collection of Generic Data of GPS Signal	2025-03-15	40
3	Collect spoofing and jamming signals	2025-03-30	60
4	Spoofing and jamming signals data argumentation	2025-04-15	160
5	Data conversion and the pre-processing, include data balancing	2025-04-30	180
6	Train and evaluate model with traditional machine learning	2025-05-20	200
7	Train the enhanced model with machine learning	2025-06-01	260
8	Fine-tune model and optimization	2025-06-25	140
9	Future work (formulate solutions for limitation)	2025-07-04	40
10	Compile and polish Final Report	2025-07-14	80
			Total: 1200

Deliverables

Items

1	A Deliverable Detection Model
2	A Deliverable Dataset with location attribute transferred to existing GPS data
3	A Publishable Paper
4	A Deliverable interface app system website for demo or video play

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